A quiescent black hole owes its existence to gravitational nonlinearity; without the gravitational nonlinearity, the hole could not hold itself together, just as without gaseous nonlinearities, the great red spot on the planet Jupiter could not hold itself together. When the imploding star that creates a black hole disappears through the hole's horizon, the star loses its ability to influence the hole in any way; most important, the star's gravity can no longer hold the hole together. The hole then continues to exist solely because of gravitational nonlinearity: The hole's spacetime curvature continuously regenerates itself nonlinearly, without the aid of the star; and the self-generated curvature acts as a nonlinear "glue" to bind itself together.

The quiescent black hole whets our appetites to learn more. What other phenomena can gravitational nonlinearity produce? Some answers may come from monitoring and decoding the ripples of spacetime curvature produced by coalescing black holes. We there might see chaotic, bizarre behaviors that we never anticipated.

To achieve this understanding will require monitoring the holes' symphonic ripples of curvature. How can they be monitored? The key is the physical nature of the curvature: Spacetime curvature is the same thing as tidal gravity. The spacetime curvature produced by the Moon raises tides in the Earth's oceans (Figure 10.3a), and the ripples of spacetime curvature in a gravitational wave should similarly raise ocean tides (Figure 10.3b).

General relativity insists, however, that the ocean tides raised by the Moon and those raised by a gravitational wave differ in three major ways. The first difference is propagation. The gravitational wave's tidal forces (curvature ripples) are analogous to light waves or radio waves: They travel from their source to the Earth at the speed of light, oscillating as they travel. The Moon's tidal forces, by contrast, are like the electric field of a charged body. Just as the electric field is attached firmly to the charged body and the body carries it around, always sticking out of itself like quills out of a hedgehog, so also the tidal forces are attached firmly to the Moon, and the Moon carries them around, sticking out of itself in a never-changing way, ready always to grab hold of and squeeze and stretch anything that comes into the Moon's vicinity. The Moon's tidal forces squeeze and stretch the Earth's oceans in a way that seems to change every few hours only because